

# The Inverse-GCD Operator $Q_N$ (Q6): Definitions, a Restricted Rayleigh Bound, and an Exploratory Spectral Programme

*Crown-jewel operator note — what is proved and what is open*

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Corrected preprint — August 2026

**Scope.** Q6 is developed here as a spectral operator on  $\{1, \dots, N\}$ . Proved: Bridge\* on single prime pairs, multi-representation Goldbach vectors, and the nonnegative cone. False as stated: universal full-spectrum floors  $\lambda_{\min} > -1/2$ . No Navier–Stokes claim.

## Status

| Claim                                                                | Status           |
|----------------------------------------------------------------------|------------------|
| Definitions of $Q_N$ , $\tilde{Q}_N$ , $H_N$                         | <b>Locked</b>    |
| Bridge*: $R(e_p - e_q) > -1/2$ on $\tilde{Q}_N$                      | <b>Proved</b>    |
| Positive orthant: $v \geq 0 \Rightarrow v^\top \tilde{Q}_N v \geq 0$ | <b>Proved</b>    |
| $\lambda_{\min}(Q_N) > -1/2$ for all $N$                             | <b>False</b>     |
| $\lambda_{\min}(\tilde{Q}_N) > -1/2$ for all $N$                     | <b>False</b>     |
| Dark-state $\Leftrightarrow$ Goldbach (June 2026 draft)              | <b>Withdrawn</b> |
| Multi-representation Bridge*                                         | <b>Proved</b>    |

## Abstract

We give a corrected account of the inverse-GCD operators used in earlier “Q6” drafts. Write

$$Q_N(i, j) = \frac{1}{\gcd(i, j)}, \quad \tilde{Q}_N(i, j) = \frac{1}{\gcd(i, j)\sqrt{ij}}, \quad 1 \leq i, j \leq N,$$

and let  $H_N$  be the degree-normalized form of  $\tilde{Q}_N$ . We prove Bridge\* for single prime pairs  $v = e_p - e_q$  and for multi-representation vectors  $v_k = \sum_{p \in \mathcal{R}(k)} (e_p - e_{k-p})$  (even  $k$ ): both satisfy  $R > -1/2$  on  $\tilde{Q}_N$ . The multi-rep case reduces to the pair case by a nonnegative cross-term identity. We also record that every nonnegative vector has nonnegative Rayleigh quotient on  $\tilde{Q}_N$ . We withdraw the full-spectrum Bridge claim  $\lambda_{\min} > -1/2$ : already for moderate  $N$ ,  $\lambda_{\min}(Q_N) < -1/2$  and  $\lambda_{\min}(\tilde{Q}_N) < -1/2$ . Earlier dark-state / Goldbach packaging and any Navier–Stokes claim from these matrices are withdrawn.

# 1 Operators

**Definition 1.1** (Raw inverse-GCD).

$$Q_N(i, j) = \frac{1}{\gcd(i, j)}, \quad 1 \leq i, j \leq N.$$

**Definition 1.2** (Normalized inverse-GCD).

$$\tilde{Q}_N(i, j) = \frac{1}{\gcd(i, j)\sqrt{ij}}.$$

In particular, for distinct primes  $p, q \leq N$ ,

$$\tilde{Q}_N(p, p) = \frac{1}{p^2}, \quad \tilde{Q}_N(p, q) = \frac{1}{\sqrt{pq}}.$$

**Definition 1.3** (Degree-normalized form). Let  $d_i = \sum_{j=1}^N \tilde{Q}_N(i, j)$  and  $D = \text{diag}(d_1, \dots, d_N)$ . Set

$$H_N = D^{-1/2} \tilde{Q}_N D^{-1/2}.$$

[Hygiene] These three matrices are different. A spectral statement about one does not automatically transfer to another, nor to any fluid commutator called “Theorem H” in companion notes.

[Withdrawn identity] Some June 2026 drafts asserted  $1/n = \sum_{d|n} \mu(d)\varphi(d)/d^2$ . This fails already at  $n = 2$  ( $1/2$  versus  $3/4$ ). No argument in this note uses that identity.

## 2 What fails: full-spectrum floors

**Proposition 2.1** (Full-spectrum Bridge is false). *There exist  $N$  such that*

$$\lambda_{\min}(Q_N) < -1/2 \quad \text{and} \quad \lambda_{\min}(\tilde{Q}_N) < -1/2.$$

*In particular the universal claims  $\lambda_{\min}(Q_N) > -1/2$  and  $\lambda_{\min}(\tilde{Q}_N) > -1/2$  for all  $N$  are false.*

*Numeric certificate.* Direct eigendecomposition yields, for example,  $\lambda_{\min}(Q_{10}) \approx -1.90 < -1/2$  and  $\lambda_{\min}(\tilde{Q}_{20}) \approx -0.505 < -1/2$ . Reproducible by the accompanying script `scripts/bridge_floor_verify.py`.  $\square$

A claimed universal lower bound  $\lambda_{\min}(H_N) \geq -3/14$  is likewise not uniform in small  $N$ ; see the companion verification suite. We do not package any  $H_N$  floor as a theorem here.

## 3 Bridge\*: single prime pair

**Theorem 3.1** (Bridge\*, single pair). *Let  $p \neq q$  be primes and  $N \geq \max(p, q)$ . For  $v = e_p - e_q$ ,*

$$R(v) := \frac{v^\top \tilde{Q}_N v}{\|v\|_2^2} = \frac{1}{2} \left( \frac{1}{p^2} + \frac{1}{q^2} \right) - \frac{1}{\sqrt{pq}} > -\frac{1}{2}.$$

*Proof.* Since  $\|v\|_2^2 = 2$ ,

$$v^\top \tilde{Q}_N v = \tilde{Q}_N(p, p) + \tilde{Q}_N(q, q) - 2\tilde{Q}_N(p, q) = \frac{1}{p^2} + \frac{1}{q^2} - \frac{2}{\sqrt{pq}},$$

hence

$$R = \frac{1}{2} \left( \frac{1}{p^2} + \frac{1}{q^2} \right) - \frac{1}{\sqrt{pq}}.$$

Then

$$R + \frac{1}{2} = \frac{1}{2} + \frac{1}{2p^2} + \frac{1}{2q^2} - \frac{1}{\sqrt{pq}} > \frac{1}{2} - \frac{1}{\sqrt{pq}}.$$

Distinct primes force  $pq \geq 6$ , so  $1/\sqrt{pq} \leq 1/\sqrt{6} < 1/2$ . Therefore  $R + 1/2 > 0$ .  $\square$

The quotient may be negative:  $(p, q) = (2, 3)$  gives  $R \approx -0.228 > -1/2$ . The bound is a *restricted* floor on difference vectors of single prime pairs, not a full-spectrum eigenvalue floor.

**Corollary 3.2** (Uniform pair floor). *For distinct primes  $p, q$ ,  $R(e_p - e_q) > -1/\sqrt{pq} \geq -1/\sqrt{6} \approx -0.408$ .*

*Proof.* From Theorem 3.1,  $R + 1/2 > \frac{1}{2} - 1/\sqrt{pq}$ , hence  $R > -1/\sqrt{pq}$ . Distinct primes give  $pq \geq 6$ .  $\square$

## 4 Positive orthant

**Proposition 4.1** (Nonnegative cone). *If  $v \in \mathbb{R}^N$  satisfies  $v_i \geq 0$  for all  $i$ , then  $v^\top \tilde{Q}_N v \geq 0$ .*

*Proof.* Every entry of  $\tilde{Q}_N$  is positive, so the quadratic form on the nonnegative orthant is a sum of nonnegative terms.  $\square$

## 5 Earlier drafts — corrected status

Earlier public drafts pushed Q6 toward Strong Goldbach via a dark-state lemma and a full-spectrum Bridge floor, and toward RH / Navier–Stokes packaging. Those extensions are not supported by the arguments as written; they are marked exploratory-failed below and are not part of the proved core.

1. **Broken detector.** The indicator difference  $v_k(j) = 1_{\mathbb{P}}(j) - 1_{\mathbb{P}}(k - j)$  vanishes on genuine Goldbach pairs  $p + (k - p) = k$ , so it does not detect them.
2. **False sign lemma.** On raw  $Q_N$ , for  $v = e_p - e_q$ ,  $\langle v, Q_N v \rangle = 1/p + 1/q - 2$  is typically negative (e.g.  $(3, 7) \mapsto -1.52$ ). A claimed equivalence “dark state  $\Leftrightarrow$  no Goldbach” does not hold as written.
3. **No NS map.** A spectral floor on  $Q_N$ ,  $\tilde{Q}_N$ , or  $H_N$  is not, by itself, a bound on  $\|(u \cdot \nabla)u\|$  for Leray–Hopf solutions on  $\mathbb{T}^3$ .

Older DOIs may remain as dated history. The public credit for this operator is Theorem 3.1, Theorem 6.2, and Propositions 2.1–4.1.

## 6 Bridge\*: multi-representation

**Lemma 6.1** (Cross-term factorization). *For distinct primes  $p \neq q$  and  $r \neq s$ ,*

$$\tilde{Q}_N(p, r) - \tilde{Q}_N(p, s) - \tilde{Q}_N(q, r) + \tilde{Q}_N(q, s) = \left(p^{-1/2} - q^{-1/2}\right)\left(r^{-1/2} - s^{-1/2}\right).$$

*In particular, if  $p < q$  and  $r < s$ , the cross term is strictly positive.*

*Proof.* Each off-diagonal entry is  $(\cdot)^{-1/2}$  for distinct primes, so the left-hand side factors as claimed.  $\square$

**Theorem 6.2** (Bridge\*, multi-representation). *Let  $k \leq N$  be even and*

$$\mathcal{R}(k) = \{p \text{ prime} : 2 \leq p < k/2, k - p \text{ prime}\}.$$

*Set  $v_k = \sum_{p \in \mathcal{R}(k)} (e_p - e_{k-p})$  (indices  $\leq N$ ). If  $v_k \neq 0$ , then*

$$R(v_k) = \frac{v_k^\top \tilde{Q}_N v_k}{\|v_k\|_2^2} > -\frac{1}{2}.$$

*Proof.* Distinct unordered Goldbach pairs for fixed  $k$  are pairwise disjoint as sets of prime sites: a shared prime forces the same complementary prime. Write  $v_k = \sum_{a=1}^m v_a$  with  $v_a = e_{p_a} - e_{q_a}$ ,  $p_a < q_a = k - p_a$ . Then  $\|v_k\|_2^2 = \sum_a \|v_a\|_2^2 = 2m$  and

$$v_k^\top \tilde{Q}_N v_k = \sum_a v_a^\top \tilde{Q}_N v_a + 2 \sum_{a < b} v_a^\top \tilde{Q}_N v_b.$$

Each diagonal block equals  $\|v_a\|_2^2 R(v_a)$  with  $R(v_a) > -1/2$  by Theorem 3.1. Each cross block equals the factorization in Lemma 6.1 with  $p_a < q_a$  and  $p_b < q_b$ , hence is positive. Therefore

$$R(v_k) \geq \frac{\sum_a \|v_a\|_2^2 R(v_a)}{\sum_a \|v_a\|_2^2} \geq \min_a R(v_a) > -\frac{1}{2}.$$

$\square$

The worst multi-rep Rayleigh through  $N = 200$  coincides with a single-pair case (e.g.  $k = 8$ , pair  $(3, 5)$ ,  $R \approx -0.183$ ). Richer multi-rep vectors sit strictly higher because of positive cross terms.

## Reproducibility

```
python3 scripts/bridge_floor_verify.py 50
python3 -m unittest tests.test_bridge_star_h_n -v
```

## Acknowledgments

This note supersedes the earlier Q6 preprint (10.5281/zenodo.20405589) as the author's intended public write-up of the inverse-GCD operator.

## References

- [1] J.R. Simons, *Earlier Q6 draft* (archive), Zenodo 10.5281/zenodo.20405589.
- [2] J.R. Simons, *Bridge\* proof draft*, repository note docs/BRIDGE-STAR-PROOF.md (2026).